

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

83000139 - Sloshing

DEGREE PROGRAMME

08IN - Master Universitario En Ingenieria Naval Y Oceanica

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	83000139 - Sloshing
No of credits	3 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	08IN - Master Universitario en Ingeniería Naval y Oceanica
Centre	08 - E.T.S. De Ingenieros Navales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Javier Calderon Sanchez (Subject coordinator)		javier.calderon@upm.es	- -
Antonio Souto Iglesias		antonio.souto@upm.es	Sin horario.

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

2.3. External faculty

Name and surname	Email	Institution
Leo Miguel González Gutiérrez	leo.gonzalez@upm.es	UPM
Daniel Duque Campayo	daniel.duque@upm.es	UPM
Pablo Eleazar Merino Alonso	peleazar.merino@upm.es	Coremarine Solutions

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

- Ampliación De Matemáticas
- Hidrodinámica Avanzada Del Buque

3.2. Other recommended learning outcomes

- Mecánica
- Mecánica de Fluidos

4. Skills and learning outcomes *

4.1. Skills to be learned

CE2-(K1) - Conocimiento avanzado de la hidrodinámica naval para su aplicación a la optimización de carenas, propulsores y apéndices. Advanced knowledge of naval hydrodynamics for its application to the optimization of hulls, thrusters and appendages.

CE3-(K2) - Conocimiento de la dinámica del buque y de las estructuras navales, y capacidad para realizar análisis de optimización de la estructura, de la integración de los sistemas a bordo, y del comportamiento del buque en la mar y de su maniobrabilidad. Ability to define the shipbuilding strategy and to plan and control the development of vessels

CG1 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio. Students should be able to apply their acquired knowledge and problem-solving skills in new or

unfamiliar environments within broader (or multidisciplinary) contexts related to their area of study.

CG2 - Que los estudiantes sean capaces de integrar conocimientos y enfrentarse a la complejidad de formular juicios a partir de una información que, siendo incompleta o limitada, incluya reflexiones sobre las responsabilidades sociales y éticas vinculadas a la aplicación de sus conocimientos y juicios. That students are able to integrate knowledge and face the complexity of making judgments based on incomplete or limited information, including reflections on the social and ethical responsibilities linked to the application of their knowledge and judgments.

CG3 - Que los estudiantes sepan comunicar sus conclusiones- y los conocimientos y razones últimas que las sustentan a públicos especializados y no especializados de un modo claro y sin ambigüedades. That students know how to communicate their conclusions - and the ultimate knowledge and reasons that support them - to specialized and non-specialized audiences in a clear and unambiguous manner.

CG4 - (S1) Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CTUPM01 - (S2) Creatividad. Los estudiantes deben resolver de forma nueva, original y aportando valor, situaciones o problemas en el ámbito de la ingeniería. Creativity. Students must solve situations or problems in the field of engineering in a new, original and valuable way.

CTUPM02 - (S3) Organización y planificación. Los estudiantes fijan objetivos, con la planificación y programación de actividades (tiempo y fases) y con la organización y gestión de los recursos necesarios para alcanzarlos. Organization and planning. Students set objectives, with the planning and scheduling of activities (time and phases) and with the organization and management of the resources needed to achieve them.

CTUPM04 - (S5) Uso de la lengua inglesa. Los estudiantes establecen conversaciones con nativos sin tener problemas de comunicación adicionales tanto de forma oral como escrita. Use of the English language. Students establish conversations with native speakers without having additional communication problems both orally and in writing.

CTUPM06 - (S7) Comunicación oral y escrita. Los estudiantes transmiten conocimientos y expresan ideas y argumentos de manera clara, rigurosa y convincente, tanto de forma oral como escrita, utilizando los recursos gráficos y los medios necesarios adecuadamente y adaptándose a las características de la situación y de la audiencia. Oral and written communication. Students transmit knowledge and express ideas and arguments in a clear, rigorous and convincing manner, both orally and in writing, using the necessary graphic resources and media appropriately and adapting to the characteristics of the situation and the audience.

CTUPM09 - Resolución de problemas. Los estudiantes son capaces de identificar o proponer un problema, y tienen el conocimiento sobre diferentes alternativas metodológicas y estratégicas para resolverlo. Problem solving. Students are able to identify or propose a problem, and have the knowledge about different methodological and

strategic alternatives to solve it.

CTUPM11 - Gestión de la información. Los estudiantes tienen la capacidad para buscar, seleccionar, ordenar, relacionar, evaluar, valorar información proveniente de distintas fuentes. Information management. Students have the ability to search, select, order, relate, evaluate, and assess information from different sources.

4.2. Learning outcomes

RA59 - S7 Comunicación oral y escrita.

RA57 - S2 Creatividad

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The course is structured into two main parts.

The first part covers general theoretical aspects of sloshing in rigid tanks. Topics include the governing equations, anti-roll tanks, vertical and horizontal sloshing, and the impact of sloshing on the damping of mechanical systems.

The second part introduces, through a practical approach, a computational method to simulate sloshing problems: the Smoothed Particle Hydrodynamics (SPH) method. For this purpose, students will use the AQUAplus code, developed at ETSIN.

This second part also includes a practical session at the Towing Tank of the ETSI Navales, where various experimental setups related to sloshing will be demonstrated (such as the Dam Break test, sloshing in single-degree-of-freedom systems, and anti-roll tanks).

Each student will select one of the sloshing problems presented and simulate it using SPH, comparing their results with existing references in the literature and other previously derived results.

5.2. Syllabus

1. Introduction: motivation and applications to naval architecture
2. Linear Potential approach
3. Modal analysis in rigid tanks
4. Experimental approaches to sloshing
5. Smoothed Particle Hydrodynamics for sloshing
 - 5.1. Governing equations
 - 5.2. Numerical model
 - 5.3. Preliminary tests: standing wave
 - 5.4. Application to sloshing

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Introduction Duration: 02:00 Lecture			
2	Linear Approximation Duration: 02:00 Lecture			
3	Linear Approximation Duration: 02:00 Problem-solving class			
4	Modal analysis Duration: 02:00 Lecture			
5	Experimental approach to sloshing Duration: 01:00 Lecture	Experiments Duration: 01:00 Laboratory assignments		Linear potential approach Individual work Progressive assessment and Global Examination Not Presential Duration: 01:00
6	Introduction to SPH Duration: 02:00 Lecture			
7	Governing equations & Numerical model Duration: 02:00 Lecture			
8		Preliminary test case Duration: 02:00 Problem-solving class		
9		Preliminary test case Duration: 02:00 Problem-solving class		
10		Application of SPH to sloshing Duration: 02:00 Problem-solving class		Standing wave results report Individual work Progressive assessment and Global Examination Not Presential Duration: 01:00
11		Application of SPH to sloshing Duration: 02:00 Problem-solving class		
12		Application of SPH to sloshing Duration: 02:00 Problem-solving class		

13		Application of SPH to sloshing Duration: 02:00 Problem-solving class		
14		Application of SPH to sloshing Duration: 02:00 Problem-solving class		
15		Application of SPH to sloshing Duration: 02:00 Problem-solving class		
16		Application of SPH to sloshing Duration: 02:00 Problem-solving class		Numerical model & Preliminary results Individual work Progressive assessment and Global Examination Not Presential Duration: 01:00
17				Control Written test Progressive assessment and Global Examination Presential Duration: 00:30 Oral presentation Individual presentation Progressive assessment and Global Examination Presential Duration: 01:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
5	Linear potential approach	Individual work	No Presential	01:00	10%	5 / 10	CG1 CG2 CG4 CTUPM09
10	Standing wave results report	Individual work	No Presential	01:00	10%	5 / 10	CG3 CTUPM04 CTUPM09
16	Numerical model & Preliminary results	Individual work	No Presential	01:00	20%	5 / 10	CG1 CG4 CG3 CTUPM04 CTUPM02 CTUPM06 CTUPM09 CTUPM11
17	Control	Written test	Face-to-face	00:30	15%	3 / 10	CE2-(K1) CG1 CG2 CG4 CG3 CTUPM04
17	Oral presentation	Individual presentation	Face-to-face	01:00	45%	5 / 10	CE3-(K2) CG1 CTUPM04 CTUPM01 CTUPM06 CTUPM11

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
5	Linear potential approach	Individual work	No Presential	01:00	10%	5 / 10	CG1 CG2 CG4 CTUPM09

10	Standing wave results report	Individual work	No Presential	01:00	10%	5 / 10	CG3 CTUPM04 CTUPM09
16	Numerical model & Preliminary results	Individual work	No Presential	01:00	20%	5 / 10	CG1 CG4 CG3 CTUPM04 CTUPM02 CTUPM06 CTUPM09 CTUPM11
17	Control	Written test	Face-to-face	00:30	15%	3 / 10	CE2-(K1) CG1 CG2 CG4 CG3 CTUPM04
17	Oral presentation	Individual presentation	Face-to-face	01:00	45%	5 / 10	CE3-(K2) CG1 CTUPM04 CTUPM01 CTUPM06 CTUPM11

7.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

7.2. Assessment criteria

The assessment will include a control test that will take place after the laboratory sessions have been completed. It will account for 15% of the final grade, with a minimum required score of 3 out of 10.

At the end of the course, a project covering the topics addressed throughout the course will be evaluated. Intermediate reports will be handled to assess that the student is following the course successfully.

The extraordinary assessment will consist of a written exam and the submission of the project and the reports.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Faltinsen & Thimoka	Bibliography	
Ibrahim - Liquid sloshing dynamics	Bibliography	
AQUAgnuSPH site	Web resource	

9. Other information

9.1. Other information about the subject

Please note that the schedule is indicative. The towing tank lab sessions will be carried out in groups, and a report must be prepared for each session. These sessions include a pre-lab quiz based on the practice guide to be used.